

Open-Set Bat Face Recognition: Progress Report

Sagi Levi lesagi@gmail.com June 2026

Goal. Build a non-invasive system that identifies individual fruit bats from facial images, supporting long-term behavioural studies without physical tagging. The system handles *open-set* scenarios: bats seen at test time were never observed during training.

Current status. A three-model comparison (Siamese pair network, ArcFace, AdaFace) has been completed across two species (rousettus, mauritius) and three background conditions, using a rigorous identity-disjoint evaluation protocol. A fully automated training and reporting pipeline is operational.

1 Key Finding: Evaluation Protocol Matters

A common pitfall in animal re-identification is *identity leakage*: splitting images randomly instead of splitting identities means the same bat appears in both training and test sets, inflating results. Table 1 demonstrates the scale of this effect.

Table 1: Same model, same data — different splitting protocol.

Protocol	ROC-AUC	F ₁	Recall
Random-pair split (leaky)	0.982	0.987	0.986
Identity-disjoint split (correct)	0.805	0.742	0.734

Takeaway: all results in this report use identity-disjoint splits exclusively.

2 Model Comparison

2.1 Rousettus — 12 individuals, 1 059 images (6/3/3 id split)

Table 2: Test-set results on rousettus (random background, 3 unseen test identities). Pair-family uses verification metrics (ROC-AUC, F₁); embedding-family also reports identification metrics.

Model	HPs	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	tuned	0.809	0.747	—	—	0.017
ArcFace	tuned	0.541	—	0.344	0.634	0.000
AdaFace	tuned	0.507	—	0.286	0.609	0.004
ArcFace	default	0.500	—	0.326	0.633	0.000

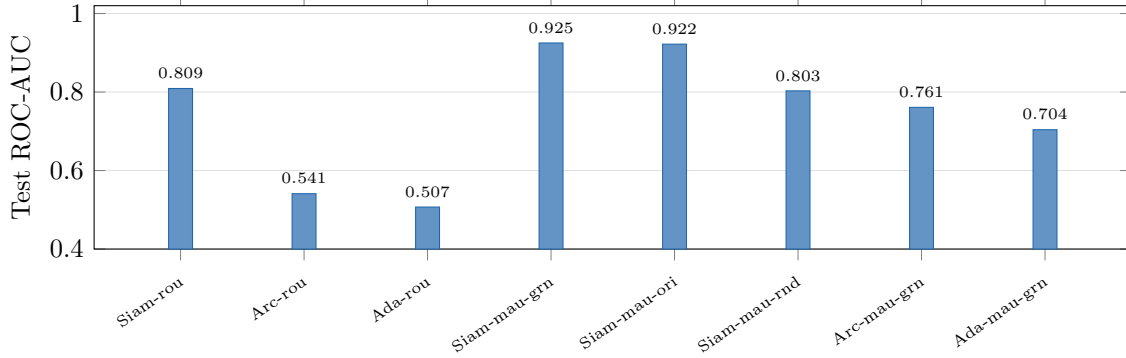
Note on ArcFace default: paper-scale hyperparameters (SGD $\eta=0.1$, margin 0.5, scale 64) destroy the ImageNet pretrained backbone on 486 training images — both train accuracy and validation ROC-AUC collapse to chance. Tuning to Adam $\eta=10^{-3}$, margin 0.2, scale 30 recovers partial performance but still falls far short of the Siamese network.

2.2 Mauritius — 11 individuals, 1 119 images (5/3/3 id split)

Table 3: Test-set results on mauritius across background conditions.

Model	Background	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	green	0.925	0.864	—	—	0.083
Siamese	original	0.922	0.872	—	—	0.041
Siamese	random	0.803	0.759	—	—	0.001
ArcFace	green	0.761	—	0.706	0.841	0.044
AdaFace	green	0.704	—	0.566	0.778	0.022
ArcFace	random	0.526	—	0.362	0.618	0.000

3 ROC-AUC Summary (bar chart)



x-axis key: Siam = Siamese, Arc = ArcFace, Ada = AdaFace; rou = roussettus (random bg); mau = mauritius with grn/ori/rnd background.

4 Why the Current Identity Count Is Insufficient

Four independent signals in the data converge on the same conclusion.

1. Train/test performance divergence. Embedding models reach near-perfect training accuracy while test ROC-AUC stays near chance—textbook closed-set overfitting:

Model	Train acc	Test ROC-AUC	Gap
ArcFace	0.988	0.541	−0.447
AdaFace	1.000	0.507	−0.493
Siamese	1.000	0.809	−0.191

2. Top-1 identification at chance. With 3 unseen test identities, random guessing gives 0.33. ArcFace scores 0.344, AdaFace 0.286—both within noise of chance. The embedding space carries no useful structure for novel identity assignment.

3. Optimization instability. ArcFace and AdaFace validation curves oscillate throughout training (val ROC-AUC jumping between 0.44 and 0.65 across consecutive epochs) and never stabilise—a symptom of an underconstrained loss landscape when too few identity classes are available.

4. Cross-species consistency isolates identity count as the cause. The same gap appears independently for both species:

Species	Train ids	Siamese	ArcFace	Gap
rousettus	6	0.809	0.541	+0.268
mauritius	5	0.803	0.526	+0.277

Had the failure been caused by species-specific factors (fur texture, pose, image quality), the gap would differ between species. Instead it is virtually identical (0.268 vs. 0.277), ruling out image-level explanations and isolating **identity count** as the governing variable.

5 Conclusions

- **Identity-disjoint evaluation is non-negotiable.** Random-pair splitting inflates F_1 from 0.74 to 0.99 on identical data.
- **Pair-based (Siamese) is the right choice at current scale.** The 0.27 ROC-AUC advantage over embedding models is confirmed independently on both species and reflects a fundamental dataset-size limitation, not a model-quality difference.
- **Background standardisation helps.** Controlled backgrounds improve Siamese ROC-AUC by ~ 12 points on mauritius.

- **Reproducibility.** All experiments tracked in MLflow; reproducible with `uv run bat-cli train -experiment <name>`.